DTC P244A:00 [SKYACTIV-D 1.5]

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DTC P244A: 00	Exhaust gas pressure sensor No.2 range/performance problem
DETECTION CONDITION	• With the following conditions met, a condition has continued for **7 s** in which the difference in the pressure before and after passing the diesel particulate filter is less than the specified value. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on) — The following conditions continue for **3 s**: • Engine speed: **825 rpm or more** • Amount of exhaust gas: **2.5 m^3/min or more** • Change in exhaust gas amount: **0.03 m^3/min or less within 0.016 s** • Exhaust gas temperature detected by exhaust gas temperature sensors No.2 and No.3: **500 °C {932 °F} or less** • Exhaust gas temperature estimated by PCM based on signal from exhaust gas temperature sensor No. 4: **500 °C {932 °F} or less** • PM accumulation amount: **1.5 g/l {0.094 lb/ft^3} or more** **Diagnostic support note** • This is an intermittent monitor (diesel particulate filter system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits the EGR control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Erratic signal to PCM — Exhaust gas pressure sensor No.2 signal malfunction — Input signal part connector or terminals malfunction — Input signal part related circuit malfunction • Exhaust gas leakage from exhaust system • Pipe between exhaust gas pressure sensor No.2 and catalytic converter (diesel particulate filter) restriction and/or damaged or improper routing • Exhaust gas pressure sensor No.2 malfunction • Catalytic converter (diesel particulate filter) or diesel particulate filter malfunction (deformation, damage) • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
4	**VERIFY CURRENT INPUT SIGNAL STATUS** • Start the engine. • Access the following PID using the M-MDS: (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) — EXHPRESS_DIF • Is the PID value within specification? (See PCM INSPECTION [SKYACTIV-D 1.5].)	Yes	Go to the next step.
		No	Inspect the suspected sensor and related wiring harness. • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results, then go to Step 8. • If there is no malfunction: — Go to the next step.
5	**INSPECT EXHAUST SYSTEM FOR LEAKAGE** • Visually inspect for exhaust gas leakage from the exhaust system. • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 8.
		No	Go to the next step.
6	**INSPECT EXHAUST GAS PRESSURE SENSOR NO.2 RELATED PIPE** • Visually inspect the exhaust gas pressure sensor No.2 related pipe for restriction, damaged and improper routing. (See EXHAUST GAS PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 8.
		No	Go to the next step.
7	**INSPECT EXHAUST GAS PRESSURE SENSOR NO.2** • Reconnect all disconnected connectors. • Inspect the exhaust gas pressure sensor No.2. (See EXHAUST GAS PRESSURE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the exhaust gas pressure sensor No.2, then go to the next step. (See EXHAUST GAS PRESSURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Catalytic converter (diesel particulate filter) or diesel particulate filter can be considered the cause. • Replace the catalytic converter (diesel particulate filter), then go to the next step. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
8	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Access the EXHPRESS_DIF PID using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Start the engine. • Race the engine at **3,500 rpm or more** for **1 min**. • Verify that the EXHPRESS_DIF monitor value is **0.2 kPa {0.002 kgf/cm^2, 0.03 psi} or more**. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
9	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.